

Sentinel Mobile Security SDK

Date: September 20, 2026

PROJECT	REPOSITORY	ACTIVE BRANCH	STATUS
Sentinel SDK	sentinel-sdk-ios	phase1-setup	In Progress

SUMMARY OF WORK COMPLETED TODAY

SETUP & BRANCHING Repository Architecture & Branch Workflow

- Initialized multi-platform project repository on GitHub (codebuilder0101/sentinel-sdk-ios).
- Configured structured Git branches: main (production releases), develop (central integration), and phase1-setup (active development).
- Established clean monorepo hierarchy separating native engines (ios/, android/) and wrapper modules (wrappers/react-native/, wrappers/flutter/).

STANDARDIZATION Standardized Capture Payload JSON Schema v1

- Drafted formal schema (specs/payload.schema.json) and concrete verified sample payload (specs/sample_payload.json).
- **Metadata Layer:** Captures SDK version, platform, session UUID, request identifiers, and execution duration.
- **User Data:** Validates customer full name, document/CPF check-digits, email RFC format, and E.164 phone numbers.
- **Device Telemetry:** Captures device identifiers, OS details, carrier metadata (MCC/MNC, operator name), battery stats, and jailbreak/root integrity heuristics.
- **Precise Geolocation:** High-accuracy GPS coordinates, altitude, heading, speed, and anti-spoofing flags.
- **Targeted App Screening:** Results of 50 targeted betting/gambling applications with risk classification.
- **Security & Tamper Resistance:** SHA-256 integrity hash, HMAC cryptographic signature, and replay nonce.

STORE COMPLIANCE Apple App Store & Google Play Compliance Engineering

- **Google Play Policy:** Integrated explicit <queries> element mapping in AndroidManifest.xml for targeted betting apps. Completely bypasses high-risk QUERY_ALL_PACKAGES permission to guarantee store approval.
- **Apple Sandbox Policy:** Built detection engine around canOpenURL(_) with LSApplicationQueriesSchemes declarations; documented diagnostic classification for apps without custom schemes.
- **Privacy Manifest:** Bundled PrivacyInfo.xcprivacy declaring exact data types and required API access reasons for iOS 17+ App Store compliance.

SCAFFOLDING Multi-Platform Core SDKs & Bridge Scaffolding

- **iOS Core SDK (Swift):** SPM configuration, Codable models, collector modules, unit tests, and .xcframework build automation script.
- **Android Core SDK (Kotlin):** sentinel-core library module with Kotlin serialization, collector services, and permission handling.

- **Cross-Platform Bridges:** Scaffolded decoupled **React Native** bridge (TypeScript + Native Modules) and **Flutter** plugin (Dart + MethodChannels).
- **Documentation:** Authored comprehensive ARCHITECTURE.md, COMPLIANCE.md, and INTEGRATION.md guides.

Current Workstream & Next Steps

- Refining native iOS telemetry collectors, CoreLocation accuracy timeout filters, and local binary .xcframework compilation.
- Proceeding with end-to-end telemetry capture validation across simulator and physical device targets.